

Statistics

COMPLETE FORMULA REFERENCE · CHEATSHEET

Descriptive Statistics

CORE

POPULATION MEAN

$$\mu = (\Sigma x_i) / N$$

SAMPLE MEAN

$$\bar{x} = (\Sigma x_i) / n$$

WEIGHTED MEAN

$$\bar{x}_w = \Sigma(w_i x_i) / \Sigma w_i$$

GEOMETRIC MEAN

$$G = (x_1 \cdot x_2 \cdot \dots \cdot x_n)^{(1/n)}$$

HARMONIC MEAN

$$H = n / \Sigma(1/x_i)$$

RANGE

$$R = x_{\max} - x_{\min}$$

IQR

$$IQR = Q_3 - Q_1$$

Variance & Dispersion

POPULATION VARIANCE

$$\sigma^2 = \Sigma(x_i - \mu)^2 / N$$

SAMPLE VARIANCE

$$s^2 = \Sigma(x_i - \bar{x})^2 / (n-1)$$

Bessel's correction: divides by n-1

POPULATION STD DEV

$$\sigma = \sqrt{\Sigma(x_i - \mu)^2 / N}$$

SAMPLE STD DEV

$$s = \sqrt{\Sigma(x_i - \bar{x})^2 / (n-1)}$$

COEFFICIENT OF VARIATION

$$CV = (s / \bar{x}) \times 100\%$$

STANDARD ERROR

$$SE = s / \sqrt{n}$$

MEAN ABSOLUTE DEVIATION

$$MAD = \Sigma|x_i - \bar{x}| / n$$

Shape & Position

Z-SCORE

$$z = (x - \mu) / \sigma$$

PERCENTILE (L METHOD)

$$L = (p / 100) \times n$$

SKEWNESS (PEARSON)

$$Sk = 3(\bar{x} - Md) / s$$

SKEWNESS (MOMENT)

$$g_1 = m_3 / s^3$$

KURTOSIS

$$g_2 = m_4 / s^4 - 3$$

Excess kurtosis (normal = 0)

QUARTILE DEVIATION

$$QD = (Q_3 - Q_1) / 2$$

OUTLIER FENCES

$$Q_1 - 1.5 \cdot IQR, Q_3 + 1.5 \cdot IQR$$

Probability

Probability Rules

ADDITION RULE

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

MUTUALLY EXCLUSIVE

$P(A \cup B) = P(A) + P(B)$

MULTIPLICATION RULE

$P(A \cap B) = P(A) \cdot P(B|A)$

INDEPENDENT EVENTS

$P(A \cap B) = P(A) \cdot P(B)$

CONDITIONAL PROBABILITY

$P(A|B) = P(A \cap B) / P(B)$

COMPLEMENT RULE

$P(A') = 1 - P(A)$

BAYES' THEOREM

$P(A|B) = P(B|A)P(A) / P(B)$

Discrete Distributions

BINOMIAL PMF

$P(X=k) = C(n, k) \cdot p^k \cdot (1-p)^{(n-k)}$
 $\mu = np \quad \sigma^2 = np(1-p)$

POISSON PMF

$P(X=k) = (\lambda^k e^{-\lambda}) / k!$
 $\mu = \lambda \quad \sigma^2 = \lambda$

GEOMETRIC PMF

$P(X=k) = (1-p)^{(k-1)} \cdot p$
 $\mu = 1/p \quad \sigma^2 = (1-p)/p^2$

HYPERGEOMETRIC

$P(X=k) = C(K, k)C(N-K, n-k) / C(N, n)$

NEGATIVE BINOMIAL

$P(X=k) = C(k-1, r-1)p^r(1-p)^{(k-r)}$

Continuous Distributions

NORMAL PDF

$f(x) = e^{-(x-\mu)^2 / 2\sigma^2} / (\sigma\sqrt{2\pi})$

STANDARD NORMAL

$\phi(z) = e^{-z^2 / 2} / \sqrt{2\pi}$

UNIFORM PDF

$f(x) = 1/(b-a) \text{ for } x \in [a, b]$
 $\mu = (a+b)/2 \quad \sigma^2 = (b-a)^2/12$

EXPONENTIAL PDF

$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$
 $\mu = 1/\lambda \quad \sigma^2 = 1/\lambda^2$

CHI-SQUARED

$\chi^2(k) = \text{sum of } k \text{ std normals}^2$
 $\mu = k \quad \sigma^2 = 2k$

T-DISTRIBUTION

$t = (\bar{x} - \mu) / (s/\sqrt{n})$
 $df = n - 1$

Inference

Confidence Intervals

MEAN (Σ KNOWN)

$$\bar{x} \pm z*(\sigma/\sqrt{n})$$

MEAN (Σ UNKNOWN)

$$\bar{x} \pm t*(s/\sqrt{n})$$

$$df = n - 1$$

PROPORTION

$$\hat{p} \pm z*\sqrt{[\hat{p}(1-\hat{p})/n]}$$

VARIANCE (CHI²)

$$[(n-1)s^2/\chi^2_U \text{ , } (n-1)s^2/\chi^2_L]$$

DIFF OF MEANS

$$(\bar{x}_1-\bar{x}_2) \pm t*SE_{diff}$$

COMMON Z* VALUES

$$90\% \rightarrow 1.645 \quad | \quad 95\% \rightarrow 1.96 \quad | \quad 99\% \rightarrow 2.576$$

Hypothesis Tests

ONE-SAMPLE Z-TEST

$$z = (\bar{x} - \mu_o) / (\sigma/\sqrt{n})$$

ONE-SAMPLE T-TEST

$$t = (\bar{x} - \mu_o) / (s/\sqrt{n})$$

TWO-SAMPLE T-TEST

$$t = (\bar{x}_1-\bar{x}_2) / \sqrt{(s_1^2/n_1+s_2^2/n_2)}$$

PAIRED T-TEST

$$t = \bar{d} / (s_d/\sqrt{n})$$

$$d = \text{differences}, df = n-1$$

PROP. Z-TEST

$$z = (\hat{p} - p_o) / \sqrt{[p_o(1-p_o)/n]}$$

CHI-SQUARE TEST

$$\chi^2 = \Sigma[(O-E)^2 / E]$$

$$E = (row\ total \times col\ total) / N$$

F-TEST (ANOVA)

$$F = MSB / MSW$$

Sample Size

FOR MEAN

$$n = (z* \cdot \sigma / E)^2$$

$$E = \text{margin of error}$$

FOR PROPORTION

$$n = z*^2 \cdot \hat{p}(1-\hat{p}) / E^2$$

$$\text{Use } \hat{p} = 0.5 \text{ if unknown}$$

POWER OF TEST

$$\text{Power} = 1 - \beta$$

EFFECT SIZE (COHEN'S D)

$$d = (\mu_1 - \mu_2) / \sigma_{\text{pooled}}$$

POOLED STD DEV

$$s_p = \sqrt{[(n_1-1)s_1^2+(n_2-1)s_2^2] / (n_1+n_2-2)}$$

Regression & Correlation

Linear Regression	
SIMPLE LINEAR MODEL	SUM OF SQUARES (ERROR)
$\hat{y} = b_0 + b_1x$	$SSE = \sum (y_i - \hat{y}_i)^2$
SLOPE	R-SQUARED
$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$	$R^2 = SSR / SST = 1 - SSE/SST$
INTERCEPT	ADJ. R-SQUARED
$b_0 = \bar{y} - b_1\bar{x}$	$R^2_{adj} = 1 - (1-R^2)(n-1)/(n-k-1)$
SUM OF SQUARES (TOTAL)	STANDARD ERROR OF ESTIMATE
$SST = \sum (y_i - \bar{y})^2$	$s_e = \sqrt{SSE / (n-2)}$
SUM OF SQUARES (REGRESSION)	MULTIPLE REGRESSION
$SSR = \sum (\hat{y}_i - \bar{y})^2$	$\hat{y} = b_0 + b_1x_1 + b_2x_2 + \dots + b_kx_k$

Correlation

PEARSON'S R
$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{[(n-1) \cdot s_x \cdot s_y]}$
COVARIANCE
$Cov(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{(n-1)}$
R FROM COV
$r = \frac{Cov(X, Y)}{(s_x \cdot s_y)}$
SPEARMAN'S P
$\rho = 1 - \frac{6 \sum d_i^2}{[n(n^2-1)]}$
<i>d_i = rank difference</i>
T-TEST FOR R
$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$

Combinatorics & Expected Value

Counting

PERMUTATION

$$P(n,r) = n! / (n-r)!$$

COMBINATION

$$C(n,r) = n! / [r!(n-r)!]$$

PERMUTATION (REPETITION)

$$n^r$$

COMBINATION (REPETITION)

$$C(n+r-1, r)$$

TOTAL ARRANGEMENTS

$$n! / (n_1! \cdot n_2! \cdot \dots \cdot n_k!)$$

For repeated elements

Expected Value & Variance

EXPECTED VALUE (DISCRETE)

$$E(X) = \sum x_i \cdot P(x_i)$$

VARIANCE (DISCRETE)

$$\text{Var}(X) = \sum (x_i - \mu)^2 \cdot P(x_i)$$

ALT VARIANCE

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

E(AX + B)

$$= a \cdot E(X) + b$$

VAR(AX + B)

$$= a^2 \cdot \text{Var}(X)$$

VAR(X ± Y) — INDEPENDENT

$$= \text{Var}(X) + \text{Var}(Y)$$

ANOVA ANALYSIS OF VARIANCE

GRAND MEAN

$$\bar{x}_G = \sum x_{ij} / N$$

SSB (BETWEEN)

$$SSB = \sum n_j (\bar{x}_j - \bar{x}_G)^2$$

SSW (WITHIN)

$$SSW = \sum (x_{ij} - \bar{x}_j)^2$$

MSB / MSW

$$MSB = SSB / (k-1) \quad | \quad MSW = SSW / (N-k)$$

F-STATISTIC

$$F = MSB / MSW$$

$$df_1 = k-1, df_2 = N-k$$

SST = SSB + SSW

$$\text{Total} = \text{Between} + \text{Within}$$

Key Symbols

μ, σ	Population mean, std dev
$\bar{x}, s$	Sample mean, std dev
n, N	Sample size, population size
$p, \hat{p}$	True proportion, sample prop.
H_0, H_1	Null and alternative hypothesis
α, β	Type I & II error rates
ρ, r	Pop. & sample correlation
λ	Poisson/Exp. rate parameter
df	Degrees of freedom
E	Margin of error / Expected value

💡 **Central Limit Theorem:** As $n \rightarrow \infty$, the sampling distribution of $\bar{x}$ approaches $\text{Normal}(\mu, \sigma^2/n)$, regardless of population shape. Use when $n \geq 30$.